

CONTENTS

♦ Selection Procedure	4
♦ General Instructions for the Candidates	5
♦ Who can Apply?	6
♦ How to Apply?	6
♦ Eligibility Criteria – HEC USAT-2022	7
♦ USAT Content Weightages	8
♦ Tips for Solving MCQs Correctly	9
♦ Fully Solved Original Model Paper (USAT-CS)	10

STUDY MATERIAL

Criteria and Subject Division	Subject 100% Weight	Page #
PART I: Number of Test Items (75 MCQs) Time Allowed 100 Minutes		
A: Verbal Reasoning	20 MCQs	17
A1: Analogy	05	18
A2: Synonym/ Antonym	04	32
A3: Sentence Completion	06	50
A4: Comprehension	05	66
B: Quantitative Reasoning	25 MCQs	81
B1: Arithmetic	06	81
B2: Algebra and Functions	04	115
B3: Geometry	03	122
B4: Equations	03	138
B5: Statistics	03	146
B6: Scenario Based/Mental Mathematics	06	160
C: Physics	08 MCQs	178
D: Computer Science	14 MCQs	201
E: Mathematics	08 MCQs	249
Part II: Time Allowed 40 Minutes		
Argument Based Essay Writing Skills 01 out of 03 (Either in English or Urdu)	15 Marks	275
Narrative Based Essay Writing Skills 01 out of 03 (Either in English or Urdu)	10 Marks	301

56. How can you make the selected character super-scripted in MS Word.
 (A) Ctrl + = (B) Ctrl + Shift + =✓
 (C) Alt + Ctrl + Shift + = (D) None of above
57. When typing in a word field manually, what must you press to insert the code's brackets?
 (A) Ctrl + F6 (B) Ctrl + F9✓
 (C) Alt + F11 (D) Shift + F12
58. Which key is used to increase left indent in MS Word?
 (A) Ctrl+I (B) Ctrl+M✓
 (C) Alt+I (D) F10
59. Changing the appearance of a document is called _____.
 (A) Proofing (B) Editing
 (C) Formatting✓ (D) All of above
60. To change margin settings in Word 2016, click _____ on the menu bar and then point to the Page Margins.
 (A) File (B) Layout✓
 (C) Format (D) Tools
61. If you need to change the typeface of a document, which menu will you choose?
 (A) Edit (B) View
 (C) Format✓ (D) Tools
62. Which simplifies the process of formatting text if the same formatting is required in more than one location?
 (A) Auto Text (B) Format Painter✓
 (C) Font Dialog Box (D) None of the above
63. Pressing the _____ key instructs Word to replace an AutoText entry name with the stored AutoText entry.
 (A) f1 (B) f2
 (C) f3✓ (D) f4
64. To select multiple cells, rows, or columns in a table that are not adjacent to one another, select the first cell and then hold down the _____ key while selecting the next cell, row, or column.
 (A) Alt (B) Shift
 (C) Enter (D) Ctrl✓
65. Which of the following is the latest version of MS Excel?
 (A) Excel 2016✓ (B) Excel 2017
 (C) Excel 2007 (D) Office 365
66. Which of the following button will allow you to add, delete, or change records in your Data Source?
 (A) 'Data Source' button (B) 'Edit' button
 (C) 'Edit Data Source' button✓ (D) 'Data Editing' button
67. HTTP stands for _____.
 (A) Hypertext Transfer Processor (B) Hypertext Transfer Protocol✓
 (C) High Transfer Protocol (D) None

E: MATHEMATICS

68. Find the missing number in the series:
 60 59 57 54 50 _____.
 (A) 45✓ (B) 47

- (C) 44 (D) 43
69. A speed of 45 km per hour is the same as:
 (A) 15 m/s (B) 12 m/s
 (C) 12.5 m/s✓ (D) 13 m/s
70. A rectangular room is 6m long, 5m wide and 4m high. What is the total volume of the room in cubic meters?
 (A) 24 (B) 30 (C) 120✓ (D) 240
71. 367 is how much below 400?
 (A) 31 (B) 33✓ (C) 35 (D) 32
72. $\frac{5}{6} \times \frac{2}{5} = ?$
 (A) 3 (B) $\frac{1}{3}$ ✓ (C) 1 (D) $\frac{7}{11}$
73. $18\frac{1}{6} \div 2\frac{1}{3} = ?$
 (A) $7\frac{12}{14}$ ✓ (B) $11\frac{7}{14}$ (C) 36 (D) $4\frac{1}{2}$
74. $2\frac{2}{3} \times \frac{5}{14} = ?$
 (A) $\frac{40}{24}$ (B) $\frac{20}{22}$ ✓ (C) $\frac{50}{42}$ (D) $\frac{21}{20}$
75. A bag contains 9 KG of sugar which is separated into packages containing 450 grams each. How many such packages can be made?
 (A) 24 (B) 18 (C) 20✓ (D) 16

PART-II

- NOTE:** (i) PART-II is to be attempted on separate Answer Book.
 (ii) Attempt all questions from PART-II.
 (iii) Extra attempt of any question or any part of the attempted question will not be considered.
 (iv) Candidate must write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.

■ Write a Argument Based Essay on any one of the following:

- i) Is Science a Curse or A Blessing? ii) Indiscipline Among Students
 iii) Patriotism

■ Write a Narrative Based Essay on any one of the following:

- i) My Hobby ii) My First Day At School
 iii) A Journey By Train

- (c) Accidental (d) Rice

A2: ANTONYM:

8. **OFTEN:**
(a) Quickly (b) Never
(c) Never (d) Sometimes✓
9. **REJECT:**
(a) Allow (b) Agree
(c) Refuse (d) Accept✓

A3: SENTENCE COMPLETION

10. Stop taking drugs lest you are caught.
(A) Might be caught✓
(B) Will be caught
(C) May be caught
(D) Would be caught
11. It is *all and one* to me whether he lives in Karachi or Hyderabad.
(A) All but one
(B) All one
(C) One and the same✓
(D) All or one
12. In his lecture, he dealt *about* the cause of the Gulf War.
(A) With✓ (B) On
(C) For (D) No improvement
13. It is *one and quarter hours* since Haris went away.
(A) One and quarter
(B) One and quarter hour
(C) One hours and quarter
(D) One hour and a quarter✓
14. Where politics fails, economics *may sometime succeed*.
(A) May sometimes succeeds✓
(B) May sometime succeeds
(C) Sometimes succeed
(D) Sometimes succeeds
15. He has been working *off and on* for several years to compile a dictionary.
(A) Regularly (B) Constantly
(C) On and off✓ (D) On or off

A4: COMPREHENSION

PASSAGE

"But we do not judge a cricketer so much by the runs he gets as by the way he gets them." In literature as in finance, says Washington Irving, "much paper and much poverty may co-exist." And in cricket, too many runs and much

dullness may be associated. If cricket is menaced with creeping paralysis, it is because it is losing the spirit of joyous adventure and becoming a mere instrument for compiling tables of averages. There are dull, mechanical fellows who turn out runs with as little emotion as a machine turns out pins. There is no colour, no enthusiasm, no character in their play. Cricket is not an adventure to them, it is a business. It was so with Shrewsbury. His technical perfection was astonishing; but the soul of the game was wanting in him. There was no sunshine in his play, no swift surprise of spending unselfishness. And without these things, without gaiety, daring and the spirit of sacrifice cricket is a dead thing. Now the Jam Sahib has the root of the matter in him. His play is as sonny as his face. He is not a miser hoarding up runs, but a millionaire spending them, with a splendid yet judicious prodigality. It is as though his pockets are bursting with runs that he wants to shower with his blessing upon the expectant multitude. It is not difficult to believe that in his little kingdom of Nawanagar where he has the power of life and death in his hands, he is extremely popular, for it is obvious that his pleasure is in giving pleasure.

16. What is/are true of Shrewsbury?

- I. He was a cricket player.
II. His technical knowledge about cricket was poor.
III. There was no spirit in his play.
(a) I and II (b) I and III✓
(c) Only I (d) None of these

17. The author feels that:

- (a) Technical perfection is not required in playing an enjoyable cricket
(b) He who pays cricket with adventure and enthusiasm makes it enjoyable✓
(c) He who scores a century must be a good cricketer
(d) Cricket is a monotonous game

18. Jam Sahib:

- (a) Was a splendid cricket player
(b) Lived in Nawanagar

FULLY SOLVED SAMPLE PAPER UNDERGRADUATE STUDIES ADMISSION TEST (USAT-CS)

Having 12 years (Intermediate or Equivalent Education)

PART-I:

Verbal Reasoning 20 MCQs
Quantitative Reasoning 25 MCQs
USAT-CS Subjects: 30 MCQs

Time: 100 Minutes
Marks: 75 Marks

PART-II:

Essay Writing Skills 25 Marks

Time: 40 Minutes

Instructions: Please read the following instructions carefully before attempting the paper.

- 75 MCQs are given on various topics, which carry equal marks. Each correct answer carries 1 mark. Attempt all the questions.
- In every question, four / five options are given (A, B, C, D, E), you have to select only one correct option.
- Mark your choice by filling in the appropriate circle against each question.
- Use ball point pen (black or blue) to shade / blacken the corresponding circle in the answer sheet.
- The candidates should carefully think about their answer before filling the circles on the answer sheet.
- Erasing, cutting or overwriting is not allowed. Once an answer has been given on the answer sheet, the candidate will not be permitted to change any of his/her answer in any way. All such answers will be treated as wrong.
- The candidate should not write any thing regarding answers on the question paper. All answers must be given on the answer sheet only.

PART-I

A: VERBAL REASONING

A1: ANALOGY

- Jesus is related to Christians in the same way as Zoroaster is related to?
(a) Jews (b) Parsis✓
(c) Tribals (d) Catholics
- South is related to North-West in the same way as West is related to?
(a) South-West (b) East
(c) North-East✓ (d) South
- Summit is related to Apex in the same way as Summon is related to?
(a) Court (b) Judge
(c) Witness (d) Beckon✓
- Distill is related to Whisky in the same

way as Brew is related to?

- (a) Ferment (b) Gin
(c) Beer✓ (d) Sugar
- DDT is related to Abbreviation in the same way as LASER is related to?
(a) Antithesis (b) Acronym✓
(c) Epigram (d) Epithet

A2: SYNONYM:

- STRIDENCY:**
(a) Harshness✓ (b) Flippant
(c) Consistency (d) Stress
- ACUTE:**
(a) Curious (b) Severe✓

- (c) Neither (A) nor (B) is true (d) Both (A) and (B) are true✓

19. "In literature as in finance... much paper and much poverty may co-exist." What does it mean?

- (a) Jam Sahib was rich man as he was the king of Nawanagar (b) Shrewsbury was a poor man (c) The cricketer who gets lot of runs may not play an enjoyable cricket✓ (d) None of these

20. What gives cricket its character?

- (a) The spirit of sacrifice (b) The spirit of joyous adventure (c) The gaiety daring attitude of the player (d) All of these✓

B: QUANTITATIVE REASONING

B1: ARITHMETIC

21. The chairs in the school hall can be set out in 35 equal rows or in 45 equal rows or in 105 equal rows are:

- (a) 600✓ (b) 400 (c) 40 (d) 80

22. Three bells toll after intervals of 6, 9 and 15 minutes, respectively. If they toll together at 5 p.m., when will they toll together next?

- (a) 6 : 30✓ (b) 5 : 30 (c) 6 : 45 (d) 5 : 45

23. It takes Riaz 30 minutes to mark a paper. Razi only need 25 minutes to mark a paper. If they both start marking papers at 10 : 00 AM, what is the first time they will finish marking a paper at the same time?

- a) 12 : 30 b) 12 : 45
c) 1 : 30✓ d) 12 : 25

24. Sonia buys two off-cuts of ribbon in a sale. One is 153 cm long. The other is 204 cm long. She cuts them so that she ends up with a number of pieces all the same length. What is the greatest length each piece can be?

- a) 39 b) 6
c) 17 d) 51✓

25. A farmer wants to fence a triangular field. He plans to put a fencing post in each corner and place other posts at equal distance along its sides. He wants the posts to be as far apart as possible. The sides of the field are 477 feet 2412 feet and 639 feet long. How far apart will the posts be?

- a) 18 feet b) 9 feet✓
c) 27 feet d) 159 feet

26. Which of the following is greater than p when $p = \frac{7}{11}$?

- I. $\frac{1}{p}$ II. $\frac{p+1}{p}$ III. $\frac{p+1}{p-1}$

- a) only I b) I and II only✓
c) I and III only d) II and III only

B2: ALGEBRA AND FUNCTIONS

27. A solid cylinder of brass 12cm high and 6cm in radius is melted and recasted into a right circular cone of diameter 16cm. Find the height of the cone.

- (a) 10.15cm (b) 20.25cm✓
(c) 28.75cm (d) 26.28cm

28. The slant height ℓ of a cone with radius of base r and height h is given by $\ell =$

- (a) $\sqrt{h^2 + r^2}$ ✓
(b) $\sqrt{h^2 - r^2}$
(c) $\pi r \sqrt{h^2 + r^2}$
(d) $\pi r \sqrt{h^2 - r^2}$

29. If a , b and c are the measures of the sides of a triangle, then area of triangle is

- (a) $\sqrt{s(s+a)(s+b)(s+c)}$, where $2s = a + b + c$

(b) $\sqrt{s(s-a)(s-b)(s-c)}$, where $s = \frac{a+b+c}{2}$ ✓

(c) $\sqrt{(s-a)(s-b)(s-c)}$, where $3s = a+b+c$

(d) $\sqrt{s(s-a)(s-b)(s-c)}$, where $2s = a+b+c$

30. Which of these sets is not a null set?

(a) $A = \{x \mid 6x = 24 \text{ and } 3x = 1\}$

(b) $B = \{x \mid x + 10 = 10\}$ ✓

(c) $C = \{x \mid x \text{ is a man older than 200 years}\}$

(d) $D = \{x \mid x < x\}$

B3: GEOMETRY

31. $20^\circ =$

(a) $360'$ (b) $630'$

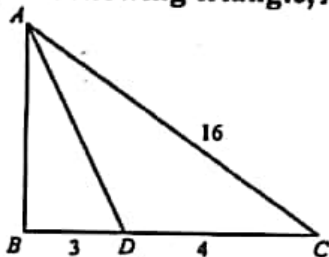
(c) $1200'$ ✓ (d) $3600'$

32. $\frac{3\pi}{4}$ radians =

(a) 115° (b) 135°

(c) 150° ✓ (d) 30°

33. In the following triangle, $AD =$



(A) $3\sqrt{2}$ (B) $6\sqrt{6}$ ✓

(C) $6\sqrt{3}$ (D) $3\sqrt{7}$

B4: EQUATIONS

34. If $3x + 17 = 9 - x$, what is the value of x ?

(A) 2 (B) 3

(C) -2 ✓ (D) -3

35. If $x - 5 = 9$, what is the value of $x^2 - 5$?

(A) 196 (B) 191 ✓

(C) 16 (D) 11

36. If $at - b = c - dt$, what is the value of t in terms of a, b, c and d ?

(A) $\frac{b-c}{a-d}$ (B) $\frac{a}{b}$

(C) $\frac{c}{d}$

(D) $\frac{b+c}{a+d}$ ✓

B5: STATISTICS

37. Primary data and ungrouped data are:

A Same

B Opposite

C Not same ✓

D Proportional

38. Statistical laws are true:

A On the

B In each case

average ✓

C In long run

D None of these

39. The data which have not undergone any statistical treatment are _____ data.

A Primary ✓

B Secondary

C Grouped

D None of these

B6: SCENARIO BASED/MENTAL MATHEMATICS

40. A tank 30 cm by 20 cm by 10 cm is $\frac{1}{5}$ full of water. How much water is in the tank?

(A) 3000 cm^3

(B) 6000 cm^3

(C) 1200 cm^3 ✓

(D) 1000 cm^3

41. During the first year, the population of a town increased by 4% and during the second year it diminished by 4%. If at the end of 2nd year, its population, was 2596, in the beginning it was:

(A) 2,550

(B) 2,600

(C) 2,400

(D) $2,500$ ✓

42. A solution of 27 gallons of acid contains 9 gallons of pure acid. How much water, in gallons, should be added to produce a 25 per cent solution of this acid?

(A) 9 ✓

(B) 18

(C) 27

(D) 25

43. At p paisa per orange, what is the price, in rupees, for 1 dozen oranges?

(A) $\frac{p}{12}$

(B) $\frac{p}{100}$

(C) $\frac{12p}{1000}$

(D) $\frac{12p}{100}$ ✓

44. Which of the following numbers belongs to the series given below: 7, 18, 29, 40

.....

(A) 2,095

(B) 2,007

(C) 2,094

(D) $2,097$ ✓

45. In a camp, there is provision for 1,600 participants to last 60 days. If the present strength of the camp is 1,200, the provision will last for _____ days.

(A) 96

(B) 86

(C) 80 ✓

(D) 72